

ALY 6010 – Probability Theory and Introductory Statistics

Final Project – Milestone 2

Week 4: Comparison Tests

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Introduction

This report represents the findings of inferential statistical analysis conducted on Statistics Canada's household Economic dataset. This dataset serves as an updated replacement for the 2023-24 survey of financial security used in the previous assignment, in response to instructor feedback recommending a more current dataset. The Household Economic Accounts dataset covers annual measures of Canadian household disposable income, final consumption expenditure and net savings from 1995 to 2025.

The dataset comprises 101,736 records across 17 variables. Three key measures namely, household disposable income, household final consumption expenditure, and household net saving were extracted and filtered to reflect national level "All household" values reported in millions of dollars, resulting in 27 annual observations across 1999 to 2025.

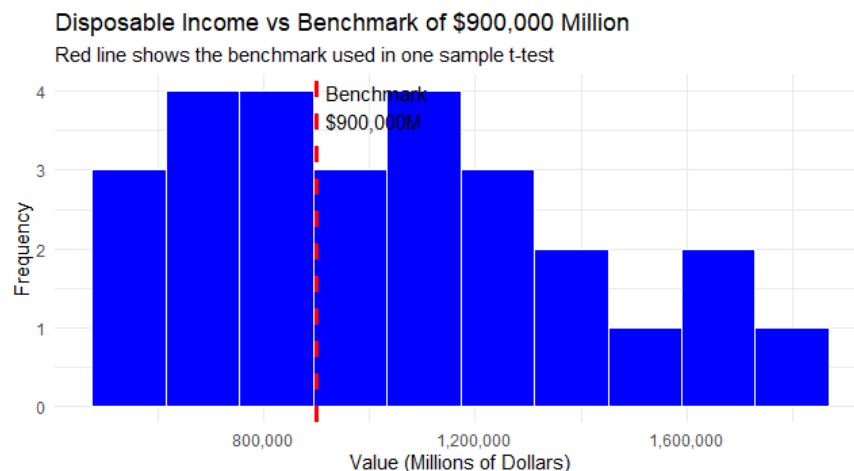
Three research questions were identified for hypothesis testing; one sample, two independent samples, and matched pairs were employed, alongside correlation and variance analysis for additional inferential measures.

Hypothesis Testing

1. Is the average Canadian household disposable income significantly different from \$900,000 million?

For this analysis, a one sample t-test was used to determine whether the mean household disposable income between 1999 and 2025 differed significantly from a benchmark or \$900,000 million, which falls near the lower end of the observed range, allowing to test whether income has grown meaningfully beyond the threshold over the period.

- $H_0: \mu = 900,000$ (mean disposable income equals \$900,000 million)
- $H_1: \mu \neq 900,000$ (mean disposable income is not equal to \$900,000 million)



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One Sample t-test

data: disposable_income$VALUE
t = 2.1084, df = 26, p-value =
0.04479
alternative hypothesis: true mean is not equal to 9e+05
95 percent confidence interval:
 903635.2 1186264.2
sample estimates:
mean of x
1044950

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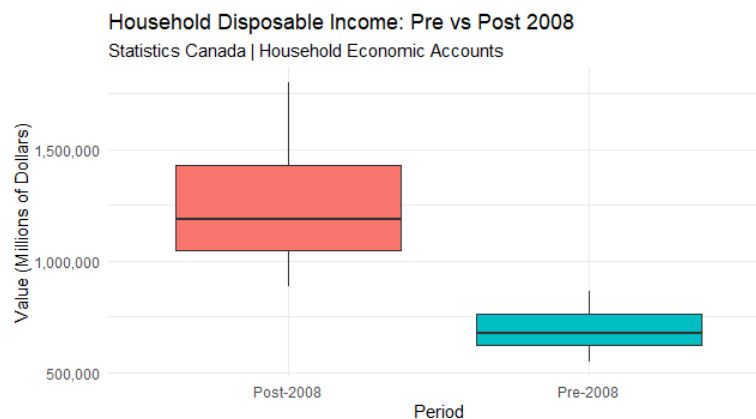
Decision: Reject H_0

Inference: Canadian household disposable income has grown significantly beyond \$900,000 million over the study period. The average income Canadians earned after 1999 was meaningfully higher than this benchmark thus confirming that household finances improved substantially over 25 years.

2. Is household disposable income significantly different between the pre-2008 and post-2008 periods?

For this analysis, a two independent samples t-test was used to compare mean household disposable incomes between pre-2008 and post-2008, two distinct economic periods, one of which represents a period of relative economic stability and another following the COVID-19 pandemic era and the global financial crisis it came with.

- $H_0: \mu_1 = \mu_2$ (no difference in mean income between pre and post 2008)
- $H_1: \mu_1 \neq \mu_2$ (significant difference exists between the two periods)



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welch Two Sample t-test

data: pre and post
t = -7.4848, df = 22.161,
p-value = 1.671e-07
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
-716450.8 -405667.9
sample estimates:
mean of x mean of y
691690.1 1252749.5

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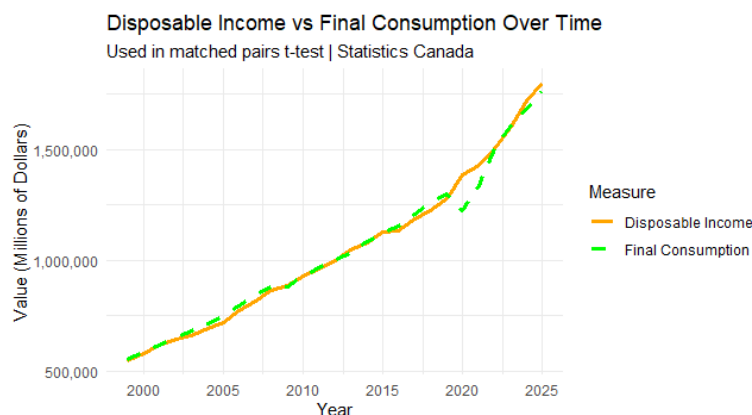
Decision: Reject H_0

Inference: Canadian households earned significantly more money after 2008 than before it. The average income nearly doubled between the two periods, from \$691,690 million to \$1,252,750 million. This reflects years of economic growth, inflation, and government spending that followed the 2008 financial crisis.

3. Is there a significant difference between household disposable income and household final consumption expenditure across the same years?

For this analysis, a matched pair t-test was used to compare disposable income and final consumption expenditure within the same years. This test was used because it observes both measures for the same time periods which makes them dependent rather than independent samples.

- $H_0: \mu_d = 0$ (no significant difference between income and consumption)
- $H_1: \mu_d \neq 0$ (significant difference exists between income and consumption)



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Paired t-test
data: disposable_income$VALUE and consumption$VALUE
t = 0.3846, df = 26, p-value = 0.7037
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
-13184.68 19254.08
sample estimates:
mean difference
3034.704
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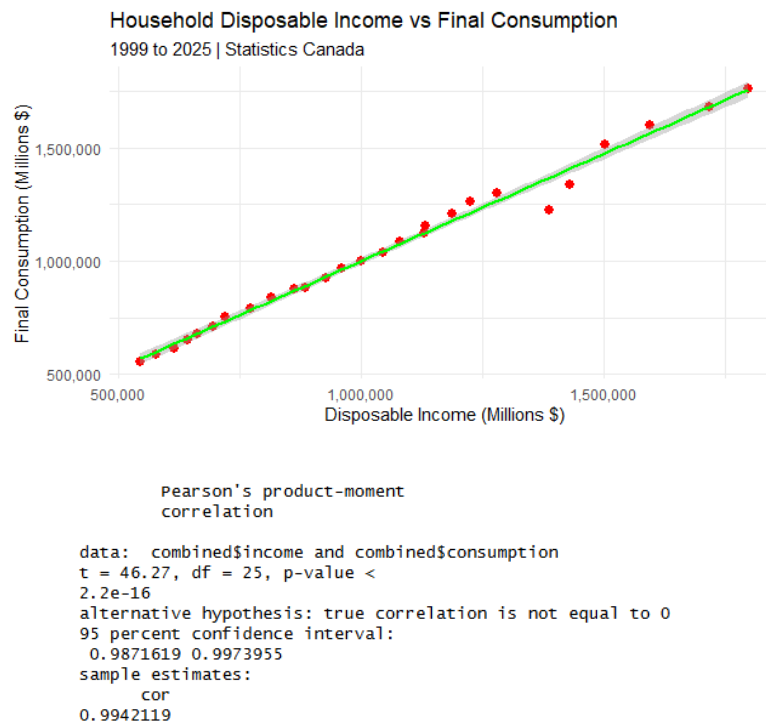
Decision: Fail to reject H_0

Inference: Canadian households spend almost everything they earn. There is no meaningful difference between how much income households receive and how much they spend in the same year, suggesting very little is left over for savings.

Correlation Analysis

A correlation analysis was conducted to identify if there is a significant linear relationship between household disposable income and final consumption expenditure and the following observations were made:

- $H_0: \rho = 0$ (no correlation between income and consumption)
- $H_1: \rho \neq 0$ (significant correlation exists)



The correlation test confirmed an extremely strong positive linear relationship between disposable income and final consumption indicating that as Canadian household income goes up, spending goes up almost perfectly in lockstep. While the two measures are so closely linked that knowing one tells you almost exactly what the other will be, it does not mean income is directly causing spending to rise (Association does not always mean causation in correlation).

A variance analysis was also conducted to identify if the variance in household disposable income significantly differs between pre-2008 and post-2008 periods.

- $H_0: \sigma_1^2 = \sigma_2^2$ (variances are equal between periods)
- $H_1: \sigma_1^2 \neq \sigma_2^2$ (variances are not equal)

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F test to compare two variances

data:  pre and post
F = 0.13636, num df = 9, denom
df = 16, p-value = 0.004697
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
 0.04472694 0.51055030
sample estimates:
ratio of variances
 0.1363614

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Decision: Reject H_0

Inference: Household income became much more unpredictable after 2008. Before 2008 income was stable and consistent year to year. After 2008 it swung more dramatically, driven by the financial crisis, rising inflation, and COVID-19 pandemic relief payments.

Conclusion

In conclusion, the analysis conducted on the dataset showed that average household disposable income reflects strong income growth over the study periods, income in the post-2008 period is significantly higher than pre-2008, the matched pairs test found no significant difference between the two measures, correlation confirmed a near perfect relationship between disposable income and final consumption, and variance test revealed that income became significantly volatile after 2008. These findings suggest that household income has grown substantially overtime, and spending has grown at an equal pace.

References

- *Distributions of household economic accounts, income, consumption and saving, Canada, provinces and Territories, annual - dataset*. Open Government Portal. (2026, April 13). <https://open.canada.ca/data/dataset/89941e0b-cc2c-43fc-a616-6e44f92d3a88/resource/72b1a523-7dd7-44bb-9dbd-7d5fdcf4dccb>